# Assertions in Python

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# What is an assertion?

An assertion is a **sanity check** that you can turn on or off when testing your program. Assertions in Python are used **to perform checks during program execution**, mainly for **debugging** purposes.

They are used to verify that a **certain condition is true** at a specific point in the code. If the **condition is not true**, Python raises an **AssertionError** exception.

**Sintax:**

```
assert condition, "Optional error message»
```

**Example:**

```
x = 5
assert x > 0, "x must be positive"
```

**Purpose of assertions:**

✅ Ensure assumptions are true during development

✅ Help detect bugs early

❌ Not for handling user input or expected runtime errors

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# The optimize (-O) parameter

If you wrote an important check using assert, that **check will never be executed** when running in **-O mode**.

**Example:**

```
assert password != "", "Password cannot be empty»
```

**Executing:**

`python -O script.py` -> assertions ignored

`python script.py` -> assertions not ignored

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